Punith Manthri

Senior React Native Engineer | Mobile Systems, JSI & Native Module Development

Bangalore, India | punithm865@gmail.com | 079753 07336 | linkedin.com/in/punithmanthri
github.com/126punith | x.com/punith_manthri | react-native-pdf-jsi

Summary

Senior React Native engineer with 5+ years designing the mobile-systems layer of enterprise Android and iOS apps in TypeScript and JavaScript. I specialize in bridging JavaScript and native code through **Native Modules, JSI, Turbo-Modules, Fabric, and Hermes**, and in the hard problems underneath the UI — **offline-first data sync, background GPS and upload services, route optimization, API batching/caching, and memory & bundle performance**.

Maintainer of open-source React Native infrastructure led by **react-native-pdf-jsi (300K+ npm downloads, 4.8/5)**, a JSI-based PDF renderer used in production. Shipped fleet and healthcare systems for **800+ drivers/agents at 99.5% uptime**, **5,000+ daily deliveries**, and users across **5+ countries** — owning architecture, native integration, CI/CD, code review, and mentoring end to end.

Core Expertise

Focus Areas: React Native Architecture · Mobile System Design · Native Module & JSI Development · BLE & Wearable Integration · Offline-First Systems · Background Processing · Performance Optimization · CI/CD

Skills

Languages: TypeScript, JavaScript, Kotlin, Java, Objective-C

Mobile: React Native, Expo, React Native CLI, Hermes, React Hooks, React Navigation, Deep Linking, Push Notifications (FCM/APNs), Bluetooth Low Energy (BLE), Wearable/Smartwatch SDK Integration

Native: Native Modules, JSI, JNI, TurboModules, Fabric, Android (Java/Kotlin), iOS (Objective-C), Gradle, CocoaPods, Android Studio, Xcode

Architecture: Mobile System Design, Offline-First, Background Services, Clean Architecture, State Management (Redux Toolkit, Context API, MobX), RBAC/Auth (OAuth, JWT)

Backend & Data: Node.js, REST APIs, GraphQL, WebSocket, MongoDB, SQLite, MMKV, AsyncStorage, Firebase, API batching & caching

DevOps & Quality: Git, GitHub Actions, CI/CD, Fastlane, App Store, Google Play, Performance Profiling, Memory & Bundle Optimization, Unit/Integration Testing

Experience

React Native Developer (Senior), Utilitarian Labs – Bangalore, India May 2025 – Sept 2026

- Led architecture of enterprise logistics and field-force apps on Android and iOS serving **800+ drivers and agents at 99.5% uptime**, owning the mobile system design, native integration, and release strategy.
- Designed a **background GPS tracking engine** on a native Android foreground service with JSI-backed batching, delivering **sub-30-second location updates** and cutting driver idle time **35%** by feeding a live route optimization layer (Google Maps).
- Architected an **offline-first trip and delivery flow** backed by local persistence and RBAC-secured REST APIs, sustaining **5,000+ daily deliveries** with reliable sync on unstable field networks.
- Delivered a **digital proof-of-delivery (POD)** pipeline with on-device capture and deferred upload that **cut paper workflows 90%** across the fleet.
- Shipped the Rockland field-agent app with geofencing, KPI dashboards, and **SAP/Salesforce integration**, lifting agent productivity **40%**.
- Owned GitHub Actions–backed release automation to the App Store and Google Play, and drove Agile/Scrum delivery through code reviews and cross-functional work with backend and product.

Software Development Engineer I, MyCLNQ Health – Bangalore, India July 2024 – May 2025

- Owned the MYCLNQ Requester healthcare appointment app end to end, shipping Android and iOS store releases to users across **5+ countries**.
- Architected a **multi-region payments layer** (Stripe, Xendit, Credimax) behind a unified interface with **OAuth/JWT-secured** transaction flows, isolating per-country provider differences from product code.
- Optimized cold start and navigation by **35%** through bundle splitting and Hermes tuning, reducing time-to-interactive for first-time users.
- Built a real-time Firebase chat with Crashlytics/Analytics observability at **99.9% uptime**, and implemented deep linking that improved retention **40%**.
- Integrated the **NerveFit smartwatch SDK over Bluetooth Low Energy (BLE)** by wrapping the native Android/iOS SDK in a **React Native native module**, handling device scanning, pairing, and auto-reconnection, and syncing health vitals from the watch into the app with **background sync**.

Frontend Developer, AMCI Edtech Pvt. Ltd – Bangalore, India Dec 2023 – May 2024

- Built the zlipp medical-education app with interactive 3D modules that lifted learner engagement **60%**.
- Designed a reusable component library that standardized UI across screens and **cut feature delivery time 30%**.

Associate Software Engineer, CognitiveClouds – Bangalore, India Sept 2022 – Oct 2023

- Delivered Sewamo (Razorpay payments, auth, store releases), WindEverest EV weather tracking, and cleanZ24 laundry with Google Maps route optimization.
- Partnered with a Rails backend on REST APIs serving **10,000+ monthly active users**, and **reduced API latency 50%** via request caching and improved data modeling.
- Contributed to code reviews and shared reusable patterns across the mobile team.

Junior Software Engineer, thesquareware – Bangalore, India Dec 2021 – Sept 2022

- Shipped the Pola and Simply-Cooked apps with Expo, offline storage (AsyncStorage), real-time chat, and role-based auth.
- Built a shared component library and patterns that increased team delivery velocity **25%**.

React Native Developer (Apprenticeship), thesquareware – Bangalore, India July 2021 – Dec 2021

- Completed a React Native apprenticeship building foundational mobile features with Redux state management and Git-based collaboration.

Independent Software Developer & Product Founder

React Native Background Location Pro 2026 – present

https://github.com/126punith/react-native-background-location-pro

- Built a **commercial background-location SDK** in **Kotlin** on React Native's **New Architecture / TurboModules**.
- Implemented **Android foreground services**, offline storage, backend ingestion, and a **live monitoring dashboard**.
- Sell via **one-time GitHub Sponsors licensing** with private source access per platform edition or bundle.
- Own architecture, documentation, release management, licensing, and developer onboarding end to end.

Open Source & Selected Projects

react-native-pdf-jsi Jan 2023 – present

High-performance PDF rendering for Android and iOS built directly on JSI native modules.

- Eliminated the React Native bridge bottleneck by binding native PDF rendering through **JSI**, achieving **~3x faster rendering** versus bridge-based alternatives.
- Maintained to **300K+ npm downloads (4.8/5)** with full TypeScript support and Hermes-friendly native integration; adopted in production document/reporting apps.

react-native-sync-vault Jan 2024 – present

Offline-first synchronization engine with SQLite persistence and network request interception.

- Designed **bi-directional sync with conflict resolution** and network/battery aware scheduling so apps stay usable and consistent while offline.
- Published and maintained on npm (**362+ downloads**) with a plugin-friendly API for custom backends.

react-native-nitro-cloud-uploader June 2023 – present

Background, resume-aware cloud upload engine with multi-provider plugins.

- Built battery-efficient **background large-file uploads** on Android/iOS with progress reporting, automatic retries, and resume-on-failure.
- Shipped a pluggable multi-provider architecture (**80K+ downloads**) so new storage backends drop in without touching core logic.

Route Optimization Engine May 2025 – present

Real-time routing and location-batching layer powering enterprise fleet tracking.

- Combined a native background GPS service with Google Maps routing to compute live route updates for **800+ drivers**, cutting idle time **35%**.
- Batched and cached location and API traffic to minimize network and battery cost at fleet scale.

Key Achievements

- **Open Source:** Scaled react-native-pdf-jsi to 300K+ npm downloads with production-grade JSI PDF rendering.
- **Enterprise Scale:** Kept fleet apps for 800+ drivers running at 99.5% uptime; digital POD cut paper workflows 90% across 5,000+ daily deliveries.
- **Native Performance:** Achieved ~3x faster PDF rendering vs bridge-based alternatives via JSI, and cut app load times 35–50% through Hermes tuning and bundle optimization.
- **Global Reach:** Expanded healthcare appointment access across 5+ countries via a multi-region payments layer and deep linking.

Education

K S School of Engineering and Management (KSSEM), BE in Mechanical Engineering – Bangalore, India Sept 2014 – June 2019

Additional Information

Languages: English (professional), Kannada (native), Telugu (proficient), Hindi (advanced)

Availability: Open to remote; immediate / short-notice join

Community: Active in the React Native community & technical writing